Nowcasting Trinidad and Tobago's Real GDP Growth

The pipeline: from 638 series to one number a month, with its band and its alarm

Jose Luis Saboin, PhD
Inter-American Development Bank

GDP Growth Nowcasting Workshop: Trinidad and Tobago

October 27, 2026

"A thousand models read the quarter; the pipeline decides whom to believe."

Roadmap

- **Part I** — Why a nowcast, and what the tool delivers
- **Part II** — The data: from 638 series to a model-ready panel
- **Part III** — The method: race, specifications, estimators, weights, regime
- **Part IV** — Identification on COVID: how the watchlist was built
- **Part V** — Production results, the live quarter, and how to operate the tool

Numbers from the production run (data through October 2026, test window 2023-Q2 to 2026-Q1, nine readings per quarter) and the identification run (test window 2019-Q4 to 2022-Q4). The companion decks cover pre-selection and the estimators: trees, LSTM, classical.

The project in five numbers

- **638 → 400:** series in the October 2026 panel, and the candidates left for the models after the filters, the redundancy rules and the panel guard — the same rules run every month.
- **6 months:** the publication delay of the official GDP print. The tool reads a quarter **nine times**, from two months before it ends to the month of the print.
- **36 groups, 170 specifications, 1,030 models:** six statistical screens propose 30 groups of indicators, a shared group, three frozen groups and two economist groups join them; each enters in up to five forms, each form is fitted by up to seven estimators; three random seeds give **3,090 readings** behind every number, at each of the nine vintages.
- **One number, one band, one alarm:** the regime-aware nowcast, the 80% and 95% bands, and a watchlist of five clusters of models that read the 2020 crash when the pool did not, from a month before the quarter closed to four months after; a cluster far above the pool is flagged, one far below it is the alarm.
- **Live, data to October:** 2026-Q2 read at $V{+}4$, $+0.1\%$ (80% band -0.9 to $+1.1$); 2026-Q3 read at $V{+}1$, -0.4% (-1.3 to $+0.6$); regime ordinary, no cluster alert.

Part I — Why a nowcast, and what the tool delivers

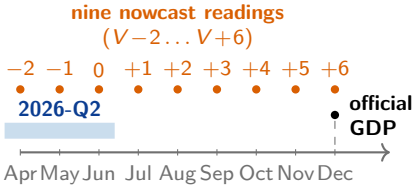
The timeliness gap

Quarterly GDP for Trinidad and Tobago is scheduled **six months** after the quarter ends (CBTT e-GDDS calendar; realised five to seven). The quarter under discussion at any policy meeting is over but not yet measured.

The tool reads that quarter **nine times**, one reading a month:

- $V-2$, $V-1$: two and one months *before* the quarter closes
- $V0$: the month it closes
- $V+1$ to $V+6$: the six months *after*, as the monthly data fill in; by $V+4$ the quarter's own months are in for 93% of the panel

The first reading lands **eight months before** the official print; the last one in the month the print is due.



Calendar for 2026-Q2: the print is scheduled for the end of December, the month of the $V+6$ reading.

What the tool delivers each month

Every monthly run writes four spreadsheets and the charts in this deck:

Regime-aware nowcast
one number per reading, weighted by past accuracy

Bands
where 80% and 95% of the weighted pool sits

Pooled median
the robust centre of all 3,090 readings

Watchlist
five clusters vs. the pooled median; the regime flag

- The **regime-aware nowcast** is the number to report. The **bands** say how much to trust it.
- The **pooled median** is the reference everything else is compared with; it is immune to a few models going wrong.
- The **watchlist** is the alarm: clusters of models that read the 2020 crash, each watched at its own month. A cluster 4 pp away from the pool is flagged; one falling 4 pp *below* it, backed by broken indicators, switches the **regime** and the weights.

Part II — The data

The panel: 638 series, October 2026 vintage

178 months, January 2012 to October 2026, in one data file and one metadata file.

- **Monetary and banking** (CBTT): loans by sector and quality, deposits, rates, liquidity, FX, reserves, financial soundness — about 300 series.
- **Fiscal** (CBTT): revenue (energy / non-energy), expenditure, financing, debt — about 145.
- **Real activity**: energy production and exports (23), cement, vehicles, card payments, labour, prices, the quarterly index of economic activity (26).
- **External**: balance of payments (38, quarterly), trade with the US, Canada, the UK and the EU, Brent, US activity (FRED).
- **Alternative data**: Google Trends (26 keywords), night-time lights by location class (13).

Frequency	Series	Share
Monthly	516	81%
Quarterly	121	19%
Publication delay		
0–2 months	137	21%
3–4 months	390	61%
5–8 months	97	15%
more than 8	13	2%

Every series carries a ticket, a description, a frequency, a publication delay, a nominal flag and an aggregation rule in the metadata. The pipeline reads nothing else.

The target: production-approach GDP, seasonally adjusted in the pipeline

- **RGDP0000**: real GDP, production approach, constant 2012 prices, 53 quarters from 2013-Q1 to 2026-Q1 (CBTT, CSO source).
- **Not seasonally adjusted at source**: raw growth averages -4.7% in first quarters and $+2.9\%$ in third quarters.
- The pipeline adjusts it with **X-13ARIMA-SEATS** (the same filter is applied to every non-adjusted indicator) and takes quarter-on-quarter growth of the adjusted level.



Raw and X-13 adjusted quarterly growth. The adjusted series still swings by about 2 pp from one quarter to the next; COVID 2020-Q2 is -12.3%.

Publication delays: measured for every series

Each series reaches the panel some months after the month it describes. That **publication delay** decides at which reading the series is visible, so it was measured for every series against release calendars and the panel itself:

Block	Series	Delay (months)	Evidence
GDP	RGDP0000	6	e-GDDS advance release calendar: timeliness 6 months; realised 5 to 7 (2026-Q1 posted 19 Oct 2026)
Index of economic activity	RIEA (26)	3	Economic DataPack, last working day of the quarter, previous quarter preliminary; realised slip to 4
Balance of payments	XBOP (38)	7	2026-Q1 values present in a panel cut in October 2026
Canada trade	XCAN (2)	2	August 2026 values present in the October panel
Discount rate	MRAT0001	1	September 2026 value present in the October panel
Energy production	RENE (17)	6–8	the October panel carries May and June values

Why it matters. At any reading, a series with a delay of d months is known only up to d months earlier; the pipeline rebuilds that view at every vintage. **The delay screen is 8 months**: series published later than that are left out; a shorter screen would drop the energy, balance-of-payments and labour blocks. A series at delay 8 still contributes at every reading, as an older observation.

From 638 series to 400 candidates: filters and rules

Standard filters (same as every country)

- at least 24 observations; at most 75% missing in the training window;
- last observation in 2024 or later;
- publication delay of at most 8 months.

Computed rules (run every time, on the levels)

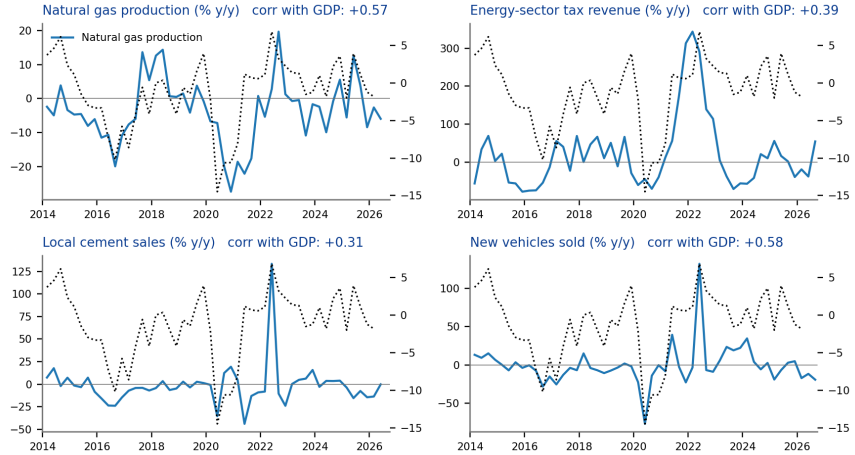
- **A0 degenerate:** constant over the last 36 observations or fewer than 10 distinct values — 51 series.
- **A duplicates:** identical series across files — 9.
- **Panel guard:** any growth rate beyond 20 standard deviations of its training window — 18 in the production run, 53 in the identification run (shorter window): rates near zero, past-due loans, debt flows.

Typed rules (from the metadata)

- **B mechanical twins:** six 30-day night-light windows, four year-on-year credit growth series, four debt ratios to GDP, the import-cover ratio — 14.
- **C identities:** components below 20% of their aggregate, in 13 ticker-block identities and four balance-of-payments ones — 17.
- **D net balances** of the balance of payments (the goods and services balances stay) — 11.
- **E by decision:** the index of economic activity (26) and one composite that ends in February 2026 — 27.

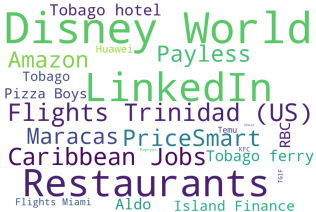
638 received → 472 after filters, A0/A and the guard → **400 candidates** (335 monthly) → the race

Structured indicators against GDP

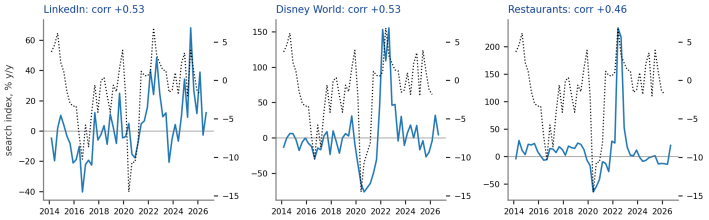


Quarterly means of the monthly series, growth over the same quarter a year earlier (left axis); GDP growth year on year, dotted (right axis). Energy drives the level, the non-energy indicators the cycle; the correlations are moderate, which is why the tool pools hundreds of them.

Alternative data (1): Google Trends



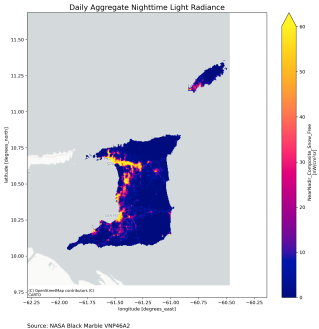
Keywords sized by their correlation with GDP (May 2026 panel).



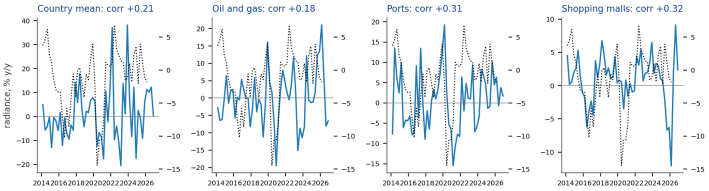
The three keywords most correlated with GDP growth on the October 2026 panel (search index, year on year; GDP dotted).

- 26 keywords on consumption (retailers, restaurants, fast food), jobs (job boards, LinkedIn) and travel (flights, Disney World, Tobago); monthly, no publication delay.
- Noisy one by one, early as a block: the pandemic shows in the search series the month it happens.

Alternative data (2): night-time lights by location class



Daily VIIRS satellite radiance (NASA), aggregated by location class.



Monthly mean radiance, year on year, against GDP growth (dotted), October 2026 panel.

- Classes: country mean, oil and gas, airports, ports, hotels, shopping malls, supermarkets — 13 series after the 30-day windows are dropped as mechanical twins.
- The oil-and-gas and mall series enter the economist groups; the rest compete in the race.

Two groups the economist picks, outside the race

Most groups of indicators come out of the **race** (Part III): six statistical screens rank the 400 candidates and propose groups of the top ones, kept if they predict GDP better than a benchmark. Two more groups are typed in from economic judgement and estimated alongside, in the same five forms and by the same estimators. They are also the **reference line** of the watchlist.

human1 — energy (10 series)

- LNG, natural gas and crude oil production; methanol and ammonia production; rig days
- Brent crude price
- night lights, oil-and-gas sites
- energy-sector tax revenue; energy exports

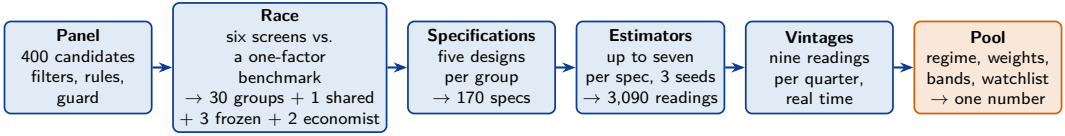
human2 — non-energy activity (14 series)

- local cement sales; new vehicle sales
- credit-card transaction value; ATM withdrawals
- private and consumer credit; deposits
- non-energy tax revenue; imports from the US and the EU
- night lights (sites mean, shopping malls); Google Trends (mean, PriceSmart)

Design: three principal components, two lags of GDP, three-month flattening, all estimators. The energy group never moved at COVID; the activity group is the reference the clusters are drawn against.

Part III — The method

The pipeline in one picture



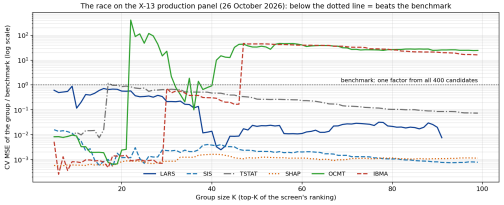
- Everything to the left of the pool is **estimation**: it runs once per re-estimation (about three hours, three seeds in parallel) and whenever the data team delivers new series.
- The pool is **reading**: it runs every month in minutes, on the readings already estimated, and decides the regime, the weights and the alarm.
- Every box is a cell of one notebook (`3_nwcst_tto.ipynb`) driven by one configuration file (`config.py`); a second notebook draws the charts.

The race: six screens, one clock to beat

Each screen ranks the 400 candidates by its own logic and proposes groups of the top K, K from 5 to 99 (the pre-selection deck works each screen by hand):

- **LARS**: order of entry into a least-angle regression
- **SIS**: sure independence screening, plain correlation
- **t-stat**: marginal regression t-statistics with GDP lags (HAC)
- **OCMT**: one-covariate-at-a-time multiple testing
- **SHAP**: importance in a boosted-tree model
- **iBMA**: iterated Bayesian model averaging

A group is scored by **expanding-window cross-validation** (17 folds): one factor of the group, predicting the quarter. It survives if it beats the **benchmark**, one factor of *all* candidates. A correlation prune at 0.8 keeps near-twins out of one group. On this panel the benchmark is very poor, so most sizes beat it: the selection is done by the cap and the spread pick (next slide).



Ratio of the group's CV MSE to the benchmark, by size, log scale. Below the dotted line the group beats the clock.

What the race picked, and the cap

Screen	Sizes beating	Best at K	Groups kept
iBMA	42	**6**	6, 17, 29, 37, 45
SIS	95	18	18, 25, 58, 80, 93
SHAP	95	18	18, 27, 47, 67, 84
OCMT	24	18	12, 18, 20, 33, 36
t-stat	94	15	15, 42, 61, 80, 99
LARS	87	42	10, 38, 42, 59, 91

"Sizes beating": how many of the 95 group sizes beat the benchmark. The single best group of all (iBMA, 6) is also kept as the *bench* specification. OCMT and iBMA select a set rather than rank everything, so their large "groups" are leftovers and blow up: only their small sizes matter.

- **Cap of five groups per screen**: without a cap the race returned 60 groups and 300 specifications. The five are spread across the winning sizes, not the five best, so the pool keeps small and large groups.
- Screens **disagree**: at $K = 60$ the overlap between any two is 13 to 42%. That disagreement is the point — it is what the pool averages over.
- **One shared group:** six variables in the largest winning group of five of the six screens (fuel imports, external debt, oil and gas output among them) form an extra static group; no variable is in all six.

Specifications: five designs, and which estimator sees which

Each group enters in up to five **designs**:

- **base**: the monthly indicators as they are
- **flats**: each indicator plus two lagged copies, so a tree sees three months at once
- **p3, p5**: three or five principal components of the group
- **p5l2**: five components plus two lags of GDP

Levels carry the signal in some groups, co-movement in others; the pool does not have to guess which.

Design	Linear	Trees	LSTM
base	ridge, ENET, LASSO	DT4, RF, GBT	yes
flats	ridge, ENET, LASSO	DT4, RF, GBT	—
p3, p5	ridge, ENET, LASSO	DT4, RF, GBT	yes
p5l2	ridge, ENET, LASSO	DT4, RF, GBT	—

The map skips cells where the design makes no difference to the estimator: the LSTM reads the monthly sequence itself, so flattening and GDP lags add nothing for it. 170 specifications (150 from the race, 5 from the shared group, 15 from the three frozen groups) plus the two economist groups, $\times$ the map $=$ 1,030 cells per seed.

Seven estimators, one line each

- **Ridge (OLSR)**: least squares with a penalty on the size of the coefficients; the penalty is chosen by time-series cross-validation. The one any economist recognises.
- **Elastic net, LASSO**: penalties that also set coefficients to zero, so the model selects within the group; penalties chosen by cross-validation (a fixed tiny penalty would be OLS in disguise).
- **DT4**: a single regression tree with leaves of at least four quarters (shallower leaves did no better in ordinary quarters; no single tree reads a crash).
- **Random forest, gradient boosting**: ensembles of trees; the most accurate families in ordinary quarters.
- **LSTM**: a small recurrent network, 20 nets of 100 epochs, reading the monthly sequence directly.
- Every tree and network cell is a **bag of 10 fits**; every cell is estimated under **three random seeds**.

The companion decks explain the three families: *Tree-based models for nowcasting*, *LSTM networks for nowcasting*, *Classical macro-econometric nowcasting*.

Vintages: every reading is real time

- For each target quarter and each of the nine readings, $V-2$ to $V+6$, the pipeline **rebuilds the data as it stood** that month: every series cut at its publication delay, GDP known only up to the last print.
- Models are trained on data **strictly before the target quarter**, then read it; next quarter they are trained again with one more print. No look-ahead anywhere, the race included.
- **Two sample splits:**
  - identification: train 2013-Q1 to 2019-Q3 (27 quarters), test 2019-Q4 to 2022-Q4 (13), so that COVID is a test quarter;
  - production: train 2013-Q1 to 2023-Q1 (41), test 2023-Q2 to 2026-Q1 (12), plus the live quarters 2026-Q2 and 2026-Q3.



The ragged edge: at each reading, each series ends where its delay puts it; the missing months are filled forward.

From 3,090 readings to one number

- **Reading guard.** A cell reading beyond ± 20 pp is dropped for that month: a few linear cells on raw designs give readings far beyond anything the target has done, and a weighted mean would follow them (69 readings dropped in the production run). No quarter has exceeded 12.3 in absolute value, so a true crash passes.
- **Pooled median.** The median of every cell's reading at that month. Robust, no weights, the reference for everything else.
- **Ordinary weights.** Each cell's weight is one over the *square* of its past mean squared error on ordinary quarters, computed on an **expanding window** of the prints known at that reading. Good cells gain weight fast; bad ones vanish.
- **Burn-in.** Until three prints are on record the weights are equal: no cell has earned anything yet. In the production window that is the first three quarters.
- **Shock weights.** When the regime says shock, the weights come from the **shock record** instead: each cell's error on past shock quarters. The record starts with the 2020 crash, read by a leave-one-out refit (next slides).
- **Bands.** Weighted percentiles of the pool: 80% (10th to 90th) and 95% (2.5th to 97.5th).

The regime: when does the tool stop trusting its ordinary record?

Two ingredients, read every month at every vintage:

- **Breadth**: how many of the monthly indicators (333 in production) are outside the range of their training window in the quarter. The generic alarm.
- **Watchlist**: five clusters of models that read the 2020 crash when the pool did not, each watched at its own vintage. Their groups are **frozen** (the lists in force at COVID, estimated beside the race in the race's own design), so the clusters do not move with the race. A cluster median 4 pp or more from the pooled median is **flagged**, on either side; only a cluster **4 pp or more below** the pool is an **alert**. The watchlist is a crash alarm; the upward flags are information.

The rule for Trinidad and Tobago: the regime switches to shock on a **cluster alert corroborated by at least five broken indicators**. Breadth alone never switches it. Once switched at a vintage, the quarter stays in shock at the later vintages. The analyst can override, and the override is logged.

Why breadth is corroboration only here.

- The alternative rule fires on breadth alone once the count passes 5% of the panel (15 of 285 in the identification run), or on a corroborated alert.
- In the identification run the count reaches 23 to 25 at COVID, but only from $V+3$ (9 to 14 before); the second lockdown reaches 16 to 18 from $V+4$; and 2022-Q4, a quarter of growth -0.4, sits at 14 for six vintages, one series short of the threshold.
- So breadth would fire three vintages after the clusters (which read COVID at $V0$) and is one series away from a false switch. It corroborates; the clusters decide.

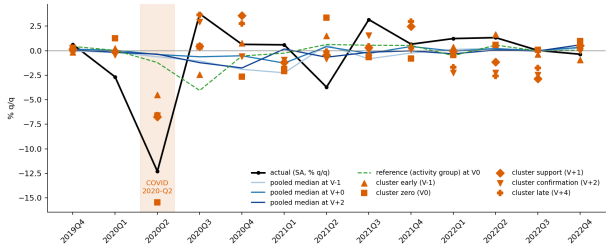
The shock record: teaching the pool what a crash looks like

- **The problem.** Ordinary weights reward accuracy in ordinary quarters. They say nothing about who reads a crash: at COVID the whole pool, weighted or not, read -0.3 to -0.6 against -12.3.
- **Leave one out.** In the production run COVID is a training quarter. For every cell the loop does one extra fit: drop 2020-Q2 from the training table, refit on the rest of the window to 2023-Q1, predict 2020-Q2. The gap to -12.3 is the cell's residual on the one shock we have.
- **Use.** When the watchlist declares a shock, the weights become one over the squared shock error. A cell that got within 6 points of the crash carries about fourteen times the weight of one that missed it by 12.
- **Honesty.** The record is one quarter of one type. The refit never sees 2020-Q2, but it does know the rebound after it, so it is fair with respect to the test window, not a June 2020 reading. The record grows with every shock quarter that later gets a print.

Part IV — Identification on COVID

Why a separate identification run

- The watchlist must be **chosen before it is used**, on a shock the models have not seen. Trinidad and Tobago has one in the sample: 2020-Q2, −12.3% seasonally adjusted.
- So the pipeline was run with the **test window starting in 2019-Q4**: COVID and the second lockdown (2021-Q2, −3.7%) are read out of sample, with 27 training quarters.
- The question for every family of cells: did it read the crash, and did it stay quiet in the twelve ordinary quarters around it?



Identification run. The pooled median misses the crash at every vintage (−0.3 to −0.6); the clusters, each at its own vintage, read −4.5 to −15.5.

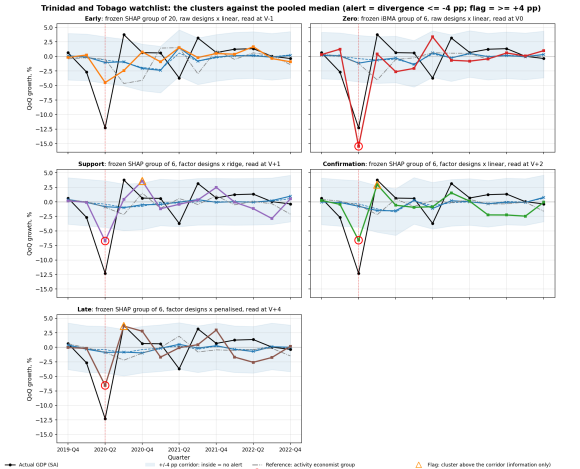
The cluster criterion and the five clusters

A cluster is a family of cells: screen × group sizes × design × estimators, read at one vintage. It qualifies if its median is **at or below −4 at COVID** and **never more than 4 pp from the pooled median** in the twelve ordinary test quarters. The search ran over every family with at least three cells at each of the nine vintages (13,743 windows, 85 qualified).

Cluster	Cells (3 seeds)	Read at	COVID	Divergence	Worst ordinary quarter
early	SHAP group of 20, raw designs, ridge / ENET / LASSO (18)	$V-1$	−4.5	−3.9 pp	−1.4 pp (2020-Q3)
zero	iBMA group of 6, raw designs, ridge / ENET / LASSO (18)	$V0$	−15.5	−15.1 pp	−2.1 pp (2020-Q4)
support	SHAP group of 6, factor designs, ridge (9)	$V+1$	−6.7	−6.3 pp	−3.0 pp (2022-Q3)
confirmation	SHAP group of 6, factor designs, ridge / ENET / LASSO (27)	$V+2$	−6.6	−6.3 pp	−2.4 pp (2022-Q3)
late	SHAP group of 6, factor designs, ENET / LASSO (18)	$V+4$	−6.6	−6.2 pp	−2.2 pp (2022-Q2)
reference	the activity economist group, all estimators (15)	all	−0.4	—	never alerts

"Worst ordinary quarter": the largest fall below the pool in the twelve other quarters; the alert line is −4 pp. The early cluster is a warning line: it read the crash a month before the quarter closed, 0.1 pp short of an alert. The race re-ranks its groups every quarter, so the clusters point at the **lists in force at COVID**, carried into production as static groups in the race's own design (base indicators, no within-quarter companions).

What the watchlist saw at COVID



- Each cluster at its own vintage against the pooled median (shaded: ± 4 pp). Circled: the alerts; hollow triangles: upward flags. **Four alerts in the whole window, all at COVID**: at $V0$, $V+1$, $V+2$ and $V+4$; none in the twelve ordinary quarters; the reference never alerts.
- With 12 to 23 broken indicators at COVID (minimum 5), the regime switches to **shock at $V0$**, the month the quarter closes, and stays switched to the print.
- **Why the alarm is one-sided.** In the rebound quarters 2020-Q3 and 2020-Q4 the three SHAP clusters read 4.2 to 4.6 pp *above* the pool (actual $+3.8$ and $+0.6$): three upward flags. Had they been alerts, the regime and the shock weights would have switched on a recovery. The clusters were identified as crash readers; an upward flag is shown, never acted on.

What the identification run taught about this panel

- **The pool does not read a crash on its own.** Pooled median -0.3 to -0.6 at every vintage against -12.3. The crash was readable, but by a few dozen cells: the iBMA group of six on raw designs at quarter end (-15.5), the SHAP group of six on factor designs from a month later. That is why the watchlist exists.
- **Breadth is a late alarm here.** Indicators outside their training range: 9 to 14 at COVID until $V+2$, 23 to 25 from $V+3$ (threshold 15); 16 to 18 at the second lockdown from $V+4$; 14 in 2022-Q4, an ordinary quarter. It confirms a shock the clusters have already read; in Trinidad and Tobago it corroborates, never switches.
- **The trees never read it.** Trees predict from leaves built on training quarters, and no training quarter looks like 2020-Q2: the best they can do is their lowest leaf. Their value is accuracy in ordinary quarters, where random forests and boosting are the best families, not shock detection.
- **The quarter after the shock is the fragile one.** Linear cells on raw designs gave readings of 40 to 60 in 2020-Q3, the rebound quarter, when the April and May values entered as features, and the crash-reading clusters leaned 4 pp above the pool. The pooled median is immune; the weighted mean is not, hence the reading guard of ± 20 pp and the one-sided alarm. Expect wide bands in the quarter after a shock.

Part V — Production results, the live quarter, and operating the tool

Accuracy in ordinary quarters, by reading

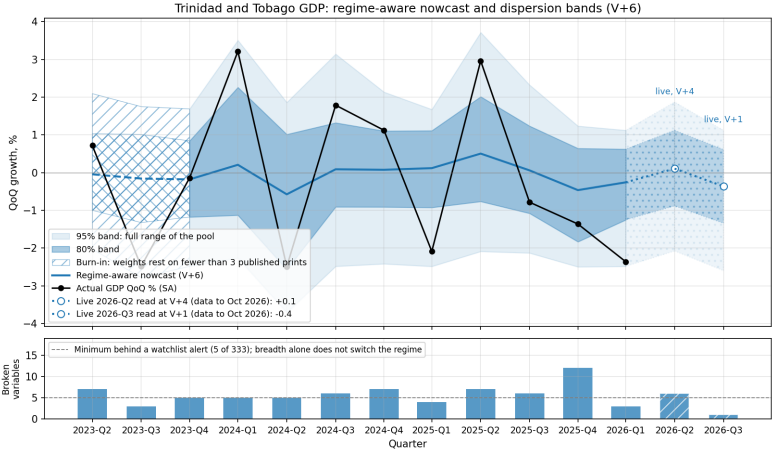
Reading	RMSE (pp)	MAE (pp)	Pooled median	80% band	95% band
$V-2$	2.15	1.98	2.11	3 of 12	9 of 12
$V-1$	1.99	1.71	2.04	5 of 12	9 of 12
$V0$	1.99	1.77	2.00	3 of 12	7 of 12
$V+1$	1.83	1.62	1.93	5 of 12	11 of 12
$V+2$	1.97	1.75	2.00	5 of 12	8 of 12
$V+3$	1.88	1.67	1.95	5 of 12	9 of 12
$V+4$	1.94	1.71	2.02	6 of 12	11 of 12
$V+5$	1.95	1.71	2.01	4 of 12	10 of 12
$V+6$	**1.82**	**1.61**	2.02	5 of 12	12 of 12
No-change forecast	3.43	—	—	—	—
Swing of the target (sd)	2.10	—	—	—	—

Twelve test quarters 2023-Q2 to 2026-Q1, no shock quarter. "Pooled median": its RMSE. Band columns: quarters in which the actual fell inside the band.

Read this honestly.

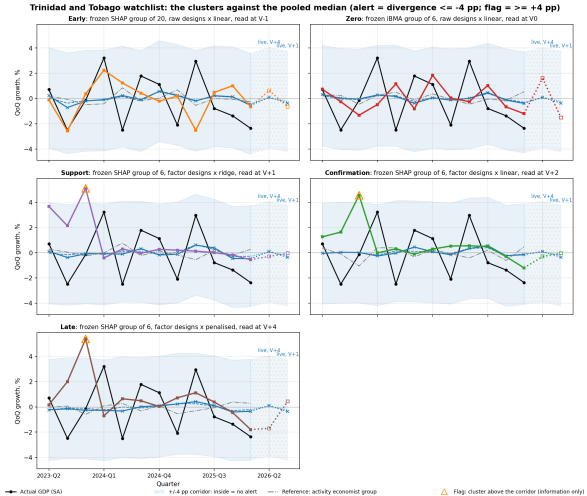
- Adjusted growth swings by ± 2.1 pp a quarter; the error is a little below the swing (1.8 to 2.2) and well below no change (3.4). **Better than no change, a little better than the mean.** With no shock in the window the weights add a tenth of a point to the pooled median.
- The **80% band is too narrow** for this target (it holds in a quarter to a half of the quarters); the **95% band holds in 7 to 12 of 12** and is the one to quote.
- **The later readings sharpen the number only a little:** from $V+1$ on the error moves between 1.8 and 2.0. What the indicators have to say about a quarter, they say by its first month after.
- In ordinary quarters the tool adds **timeliness and a band**; its distinctive value is in the quarter that is not ordinary.

The reading, with its band



- Regime-aware nowcast at $V+6$, the last reading, with the 80% band (dark) and the 95% band (the full range of the pool); **hatched:** the first two quarters, weights resting on fewer than three prints.
- The nowcast stays within ± 0.6 pp of zero while the actual swings; at $V+6$ the **95% band** catches 12 of 12 quarters, the 80% band 5 of 12. The strip below: broken indicators, never above 12 of 333; minimum behind an alert, 5.
- **Live** (data to October 2026): 2026-Q2 read at $V+4$, $+0.1$% (95% band -2.1 to $+1.9$); 2026-Q3 read at $V+1$, -0.4% (-2.6 to $+1.1$).

The watchlist in production



- The five frozen clusters (18, 18, 9, 27 and 18 cells) against the pooled median at their own vintage, with the ± 4 pp corridor; the reference line is the activity economist group.
- **No alert in the twelve ordinary quarters**: the largest fall below the pool is 2.4 pp (early, 2025-Q2). This is the out-of-sample false-alarm test of the identification: twelve quarters, zero alerts, as in the identification window.
- The three SHAP clusters read 4.5 to 5.5 pp *above* the pool in 2023-Q4 (actual -0.1): three upward flags, the same lean as in the 2020 rebound, and the production case for keeping them flags rather than alerts.
- Hollow markers: the live quarters. For 2026-Q2 at $V+4$ every cluster sits within 1.8 pp of the pool; for 2026-Q3 at $V+1$ within 1.3 pp.

The live quarters: 2026-Q2 and 2026-Q3 read in October

	2026-Q2	2026-Q3
Reading	$V+4$	$V+1$
Regime	ordinary	ordinary
Broken indicators (of 333)	6	1
Cluster alerts	none; all within 1.8 pp of the pool	none; all within 1.3 pp
Weights	ordinary, 12 prints	ordinary, 12 prints
Cells read	1,030	1,030
Regime-aware nowcast	$+0.1\%$	-0.4%
Pooled median	$+0.1\%$	-0.2%
80% band	-0.9 to $+1.1$	-1.3 to $+0.6$
95% band	-2.1 to $+1.9$	-2.6 to $+1.1$
Print due	end of December 2026	end of March 2027

How to read it.

- Two **ordinary quarters**: nothing in the indicators has broken, no cluster has fallen below the pool, so each number is the weighted average of 1,030 models trained on ordinary quarters (the LSTM has no live reading).
- Q2, with its own months all in, reads **flat**: a 95% band of -2 to $+2$ around a tenth of a percent. The zero cluster leans up by 1.6 pp, the late cluster down by 1.8; the pool sits between them.
- Q3 reads **flat to slightly negative** on the first month of data after the quarter; the 95% band says anything between -2.6 and $+1$ is consistent with the data to October.

Operating the tool

Cadence

- One run per month as the data land: nine readings per quarter. The monthly reading refreshes in minutes; a full re-estimation takes **about three hours** (three seeds in parallel).
- Re-estimate when the data team delivers new series or a new GDP vintage changes the seasonal adjustment.

Files

- `config.py`: every choice in this deck is a named parameter — sample dates, delay screen, redundancy lists, economist groups, race cap, estimators, regime rule, clusters.
- `3_nwcst_tto.ipynb`: nine cells, tagged setup to live; `run_tto.py`: the three-seed orchestrator.

What to look at each month

- The **regime flag** and the trigger behind it (watchlist, corroborated by how many broken indicators).
- The **regime-aware nowcast** with its 80% band; the pooled median beside it. If they disagree by more than the band, read the watchlist.
- The **cluster readings**: which cluster moved, at which vintage, and on which side (a flag above, an alert below).
- The **override log**: every manual regime change, with its reason.

Outputs

- Four spreadsheets (regime-aware nowcast, bands, watchlist, live quarter) and the charts of this deck.

Takeaways and open items

- **The data rules do most of the work.** Filters, computed and typed redundancy rules and the panel guard take 638 series to 400 candidates every month, with no judgement call per run.
- **Six statistical screens propose the groups, not one**, capped at five groups each, and the pool averages over their disagreement: 170 specifications, 1,030 models, three seeds, a reading guard on the pool.
- **In ordinary quarters the tool buys timeliness and a band, not a much sharper number:** error of 1.8 to 2.2 pp against 3.4 for no change and a 2.1 pp swing of the target, at every one of the nine readings; the 95% band holds in 7 to 12 of 12 quarters.
- **The watchlist is the alarm the pool lacks.** Five clusters identified on COVID, their groups frozen in the race's own design; a warning a month before the quarter closed and alerts from the month it closed to four months after, none in the twelve identification quarters nor in the twelve production quarters. The alarm is one-sided: a cluster above the pool is flagged, only one below it can switch the regime.
- **Open:** one shock on the record is the limit of the shock weights; the record grows with every shock quarter that later gets a print.

Thank you

Questions and discussion

Jose Luis Saboin, PhD — Inter-American Development Bank

GDP Growth Nowcasting Workshop: Trinidad and Tobago, October 2026

Backup material follows: the cluster definitions as configured, the redundancy lists, the full table of publication delays.

Backup — The clusters as configured

Section 7 of `config.py`. Each entry names the screen, the designs, the estimators and the range of group sizes; the post-estimation cell resolves it against the specifications of the current run.

Entry	Vintage	Screen, designs	Estimators	K	Alerts
early_vm1	-1	fshap; base, flats	OLSR, ENET, LASSO	20	yes
zero_v0	0	fibma; base, flats	OLSR, ENET, LASSO	6	yes
support_vp1	$+1$	fshap; p3, p5, p5l2	OLSR	6	yes
confirm_vp2	$+2$	fshap; p3, p5, p5l2	OLSR, ENET, LASSO	6	yes
late_vp4	$+4$	fshap; p3, p5, p5l2	ENET, LASSO	6	yes
reference	all	human; base	all	2	no

`fibma` and `fshap` are the frozen groups of section 3 (`FROZEN_GROUPS`: the iBMA top 6, the SHAP top 6 and the SHAP top 20 at the ranking in force at COVID). Other section-7 parameters: `READING_GUARD = 20`; `BREADTH_TRIGGER = False` (corroboration only); `ALERT_PP = 4`, `ALERT_SIDE = "below"`, `FLAG_BOTH = True` (a divergence on either side is flagged; only the side below alerts); corroboration minimum $\max(3, \lfloor \text{threshold}/3 \rfloor)$ broken indicators; burn-in of three prints; override dictionaries, logged.

Backup — The redundancy rules in full

B — mechanical twins (14)

- six 30-day night-light windows (UNTL3000–3006), twins of the monthly means
- four year-on-year credit growth series, twins of the levels
- four debt ratios to GDP, twins of the debt levels
- the import-cover ratio

D — signed net balances (11)

- balance-of-payments net positions (income, transfers, financial account lines); the goods and services balances are kept

C — identity components below 20% (17)

- revenue: GCGR1320, GCGR1520
- loans by sector: MLOA1102, 1105, 1302, 1305, 1401, 1402, 1405, 1501, 1502, 1504, 1505, 1705
- balance of payments: XBOP1113, XBOP1311, XBOP1321

E — by decision (27)

- the index of economic activity, 26 series (RIEA1000–1350)
- one composite series (UBML0000), ends February 2026

Detail in `data/redundancy_202610.xlsx` (sheet E_final) and `redundancy_202610.json`; lists in `config.py` section 2.

Backup — Publication delays by block

Block	Source file	Series	Delay (months)
Bank loans by sector and quality	cbtt_cb_loans	137	3
Central government operations	cbtt_cgo, cg_rev, cg_exp, cg_fin, cg_bal	111	3–4
Balance of payments (quarterly)	cbtt_bop	38	7
Financial soundness (quarterly)	cbtt_findsound	32	1–7
Monetary aggregates, deposits, FX, liquidity	cbtt_monetary, cb_deposits, cb_fx, liq, fxres	80	2–4
Google Trends	gtrends	26	0
Index of economic activity (quarterly)	cbtt_qiea	26	3 (excluded)
Energy production and exports	cbtt_prodoil	23	2–8
Interest rates	cbtt_cb_rates, repo	23	0–1
Debt	cbtt_debt_cg, debt_gg	33	3–10
Daily FX, capital market	cbtt_fx_day, capmark	31	0–2
Night-time lights	ntl_data, ntl_shape, blk_ntl	15	0
Labour (quarterly)	cbtt_lab_sec, lab_lf, lab_ur	13	7
Cards, payment terminals	cbtt_dc/cc_transactions, pay_terminals	14	5
Vehicles, cement, CPI	cbtt_vehicles, vehicles_sales, cement, cpi	12	2–4
External trade, Brent, US activity	uscensus, canada, uk, eu_trade, eia_brent, fred	16	0–3
Agricultural production	cbtt_agprod	7	14 (excluded)
GDP	cbtt_gdp	1	6

Delays as recorded in `data/meta_202610.csv`.